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Proffered Paper

Preservation of visual acuity after eye-sparing Ru-106 brachytherapy in choroidal melanoma patients

Lennart J. Pors¹, Marina Marinkovic², Heloisa H. Deuzeman¹, T.H. Khanh Vu², Ellen M. Kerkhof¹, Coen R.N. Rasch¹, Jaco C. Bleeker², Carien L. Creutzberg¹, Jan-Willem M. Beenakker^{1,2,3}, Gregorius P.M. Luyten², Nanda Horeweg^{1,2}

¹Leiden University Medical Center, Radiation Oncology, Leiden, Netherlands. ²Leiden University Medical Center, Ophthalmology, Leiden, Netherlands. ³Leiden University Medical Center, Radiology, Leiden, Netherlands

Purpose/Objective:

Uveal melanomas can be treated with enucleation or eye-sparing therapies, such as Ru-106 or I-125 plaque brachytherapy or proton beam therapy. For small tumours, Ruthenium-106 brachytherapy is a highly effective treatment option that can help preserve eyes and vision due to its small therapeutic range and steep dose falloff. In this study, we report long-term follow-up outcomes of a large cohort of patients treated with Ruthenium-106 brachytherapy and focus on the most relevant clinical outcomes: survival, local tumour control, eye preservation, and visual acuity.

Material/Methods:

All patients treated at our national referral centre for choroidal melanoma with Ruthenium-106 brachytherapy between January 2012 and March 2014 were included in our database as part of our ongoing 2012-2022 evaluation. Exclusion criteria were distant metastases at baseline and other primary treatment.

Patients were diagnosed by an ocular oncologist and screened for distant metastases. Eligibility criteria for brachytherapy were tumour prominence including sclera ≤ 7 mm and basal diameter ≤ 16 mm. Tumours invading